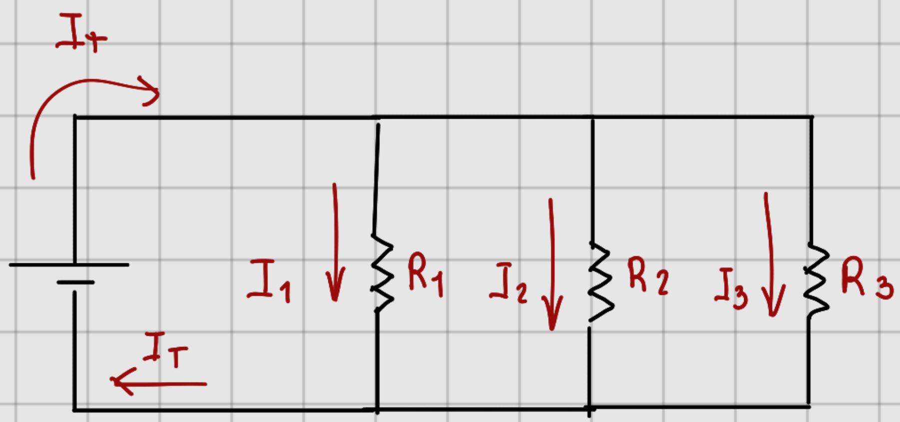

$I_T = 100mA$

$I_1 = 30mA$

$I_2 = 50mA$

$I_3 = 20mA$

$$I_T - I_1 - I_2 - I_3 = 0$$

$$100mA - 30mA - I_2 - 20mA = 0$$

$$50mA - I_2 = 0$$

$$I_2 = 50mA$$

$R_1 = 1K\Omega$

$R_2 = ?$

$R_3 = ?$

$R_T = ?$

$V = ?$

$$V_T = I_1 R_1 = 30\cancel{m}A(1\cancel{K}\Omega)$$
$$= 30V$$

$$R_2 = \frac{V_T}{I_2} = \frac{30}{50mA} = 600\Omega$$

$$R_3 = \frac{V_T}{I_3} = \frac{30}{20mA} = 1.5K\Omega$$

$$R_T = \frac{1}{1/600 + 1/1500 + 1/1000} = \frac{1}{3.333m\Omega} = 300\Omega$$

$$I_T - I_1 - I_2 - I_3 = 0$$

$$\frac{V_T}{R_T} - \frac{V_T}{R_1} - \frac{V_T}{R_2} - \frac{V_T}{R_3} = 0$$

$$\frac{30}{300} - \frac{30}{1000} - \frac{30}{600} - \frac{30}{1500} = 0$$

$$100mA - 30mA - 50mA - 20mA = 0$$

$R_1 = 1K\Omega$

$R_2 = 600\Omega$

$R_3 = 1.5K\Omega$

$R_T = 300\Omega$

$V = 30V$